

Worksheet 7.1, 7.2 and 7.3

- 7-1.** A population has a mean of 125. If a random sample of 8 items from the population results in the following sampled values, what is the sampling error for the sample?

103	123	99	107	121	100	100	99
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- 7-2.** The following data are the 16 values in a population:

10	5	19	20	10	8	10	2
14	18	7	8	14	2	3	10

- Compute the population mean.
- Suppose a simple random sample of 5 values from the population is selected with the following results:

5	10	20	2	3
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Compute the mean of this sample.

- Based on the results for parts a and b, compute the sampling error for the sample mean.

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- 7-22.** A population with a mean of 1,250 and a standard deviation of 400 is known to be highly skewed to the right. If a random sample of 64 items is selected from the population, what is the probability that the sample mean will be less than 1,325?
- 7-23.** Suppose that a population is known to be normally distributed with $\mu = 2,000$ and $\sigma = 230$. If a random sample of size $n = 8$ is selected, calculate the probability that the sample mean will exceed 2,100.

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- 7-26.** Suppose nine items are randomly sampled from a normally distributed population with a mean of 100 and a standard deviation of 20. The nine randomly sampled values are

125	95	66	116	99
91	102	51	110	

Calculate the probability of getting a sample mean that is smaller than the sample mean for these nine sampled values.

- 7-28.** Given a distribution that has a mean of 40 and a standard deviation of 13, calculate the probability that a sample of 49 has a sample mean that is
- greater than 37
 - at most 43
 - between 37 and 43
 - between 43 and 45
 - no more than 35

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- 7-33.** Suppose the life of a particular brand of calculator battery is approximately normally distributed with a mean of 75 hours and a standard deviation of 10 hours.
- What is the probability that a single battery randomly selected from the population will have a life between 70 and 80 hours?
 - What is the probability that 16 randomly sampled batteries from the population will have a sample mean life of between 70 and 80 hours?
 - If the manufacturer of the battery is able to reduce the standard deviation of battery life from 10 to 9 hours, what would be the probability that 16 batteries randomly sampled from the population will have a sample mean life of between 70 and 80 hours?

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7.3 Sampling distribution of a Proportion

7.45 A population has a proportion equal to 0.30. Calculate the following probabilities with $n=100$:

a. $P(\bar{p} \leq 0.35)$

b. $P(\bar{p} > 0.40)$

c. $P(0.25 \leq \bar{p} \leq 0.40)$

d. $P(\bar{p} < 0.27)$

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7.46 If a random sample of 200 items is taken from a population in which the proportion of items having a desired attribute is $p = 0.30$. What is the probability that the proportion of successes in the sample will be less than or equal to 0.27?

7.48 Given a population in which the proportion of items having a desired attribute is $p = 0.25$, if a sample of 400 is taken,

- a. What is the standard deviation of the sampling distribution of $\bar{p}$?
- b. What is the probability that the proportion of successes in the sample will be greater than 0.22?

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7.57 According to the most recent Labor Department data, 10% of engineers (electrical, mechanical, civil and industrial) were women. Suppose a random sample of 50 engineers is selected.

- a.** How likely is it that the random sample of 50 engineers will contain 8 or more women in these positions?
- b.** How likely is it that the random sample will contain fewer than 5 women in these positions?
- c.** If the random sample included 200 engineers, how would this change your answer to part b?